JAVA CODE AND DEMONSTRATION

QUESTION 1: OCCURRENCE OF LARGEST NUMBERS

```
import java.util.Scanner;

public final class OccurrenceOfLargestNumbers {
    private OccurrenceOfLargestNumbers() {
    }

    public static void main(String[] args) {
        try (Scanner scanner = new Scanner(System.in)) {
            System.out.print("Enter numbers: ");
            Result result = findLargestAndCount(scanner);
            System.out.println("The largest number is " + result.largest);
            System.out.println("The occurrence count of the largest number is "
+ result.count);
        }
    }

    private static Result findLargestAndCount(Scanner scanner) {
        boolean hasInput = false;
        int largest = 0;
        int count = 0;

        while (scanner.hasNextInt()) {
            int number = scanner.nextInt();
            if (number == 0) {
                break;
            }

            if (!hasInput || number > largest) {
                largest = number;
                count = 1;
                hasInput = true;
            } else if (number == largest) {
                count++;
            }
        }

        // Assumption: at least one non-zero integer is entered before the senti
nel.
        if (!hasInput) {
            throw new IllegalArgumentException("Enter at least one non-zero inte
ger before 0.");
        }
        return new Result(largest, count);
    }

    private static final class Result {
        private final int largest;
        private final int count;

        private Result(int largest, int count) {
            this.largest = largest;
            this.count = count;
        }
    }
}
```

QUESTION 1 DEMO

```
$ printf "3 5 2 5 5 5 0\n" | java OccurrenceOfLargestNumbers
Enter numbers: The largest number is 5
The occurrence count of the largest number is 4
```

QUESTION 2: CUSTOM INT WRAPPER

```
import java.util.Objects;

public final class MyInt implements Comparable<MyInt> {
    private final int value;

    public MyInt(int value) {
        this.value = value;
    }

    public int intValue() {
        return value;
    }

    public short shortValue() {
        return (short) value;
    }

    public byte byteValue() {
        return (byte) value;
    }

    public long longValue() {
        return value;
    }

    public float floatValue() {
        return value;
    }

    public double doubleValue() {
        return value;
    }

    @Override
    public int compareTo(MyInt other) {
        return Integer.compare(value, Objects.requireNonNull(other, "other").val
ue);
    }

    @Override
    public boolean equals(Object other) {
        if (this == other) {
            return true;
        }
        if (!(other instanceof MyInt)) {
            return false;
        }
        MyInt myInt = (MyInt) other;
        return value == myInt.value;
    }

    @Override
    public int hashCode() {
        return Integer.hashCode(value);
    }
```

```
    @Override
    public String toString() {
        return Integer.toString(value);
    }

    public static void main(String[] args) {
        MyInt first = new MyInt(42);
        MyInt sameValue = new MyInt(42);
        MyInt larger = new MyInt(100);

        System.out.println("Value: " + first.intValue());
        System.out.println("As short: " + first.shortValue());
        System.out.println("Equal values: " + first.equals(sameValue));
        System.out.println("Comparison with 100: " + first.compareTo(larger));
        System.out.println("String form: " + first);
    }
}


QUESTION 2 DEMO
$ java MyInt
Value: 42
As short: 42
Equal values: true
Comparison with 100: -1
String form: 42
```